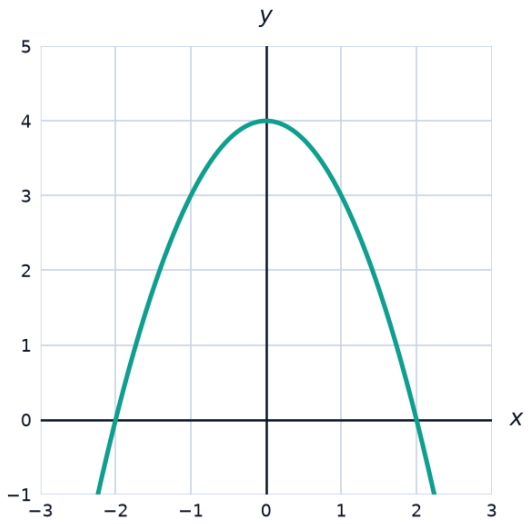
Area Between Curves

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

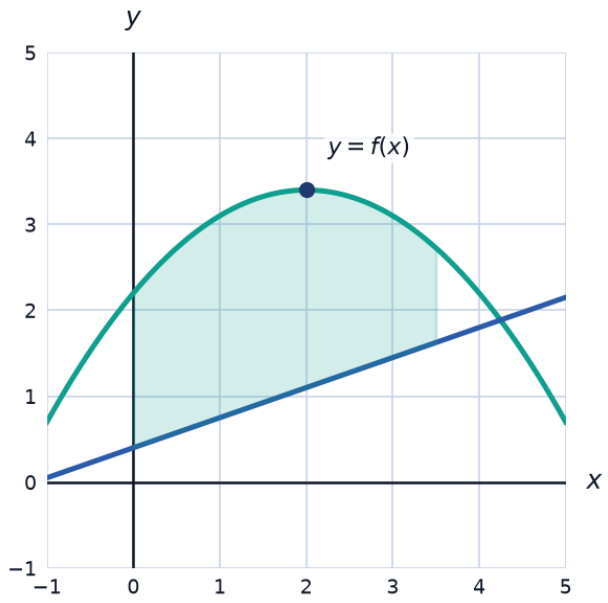
1 [Foundational] • Target: Area between a curve and the x-axis

Find the area of the region bounded by $y = 4 - x^2$ and the x-axis.



2 [Foundational] • Target: Area between two curves (vertical strips)

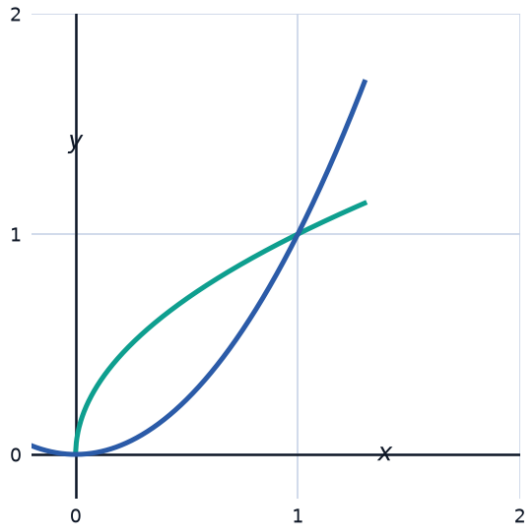
Find the area between $y = x^2$ and $y = x + 2$.



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3 [Foundational] • Target: Area between two curves (vertical strips)

Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.



4 [Foundational] • Target: Regions requiring horizontal strips

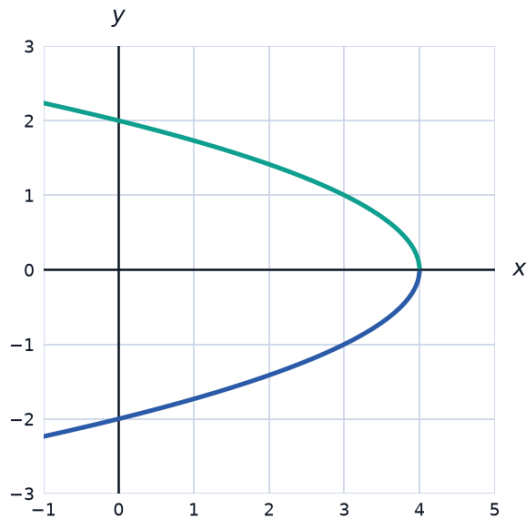
Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y.

5 [Foundational] • Target: Regions with multiple intersection points

Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.

6 [Foundational] • Target: Area using the more efficient variable of integration

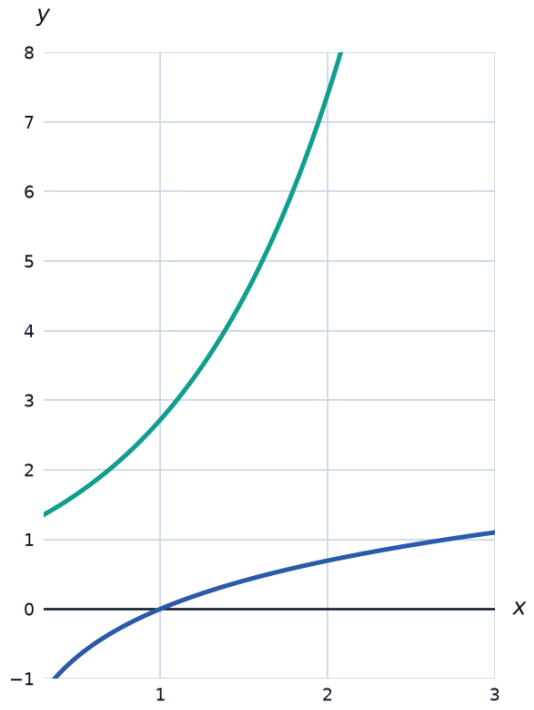
Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.



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7 [Foundational] • Target: Setting up but not evaluating (AP-style)

Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Setting up but not evaluating (AP-style)

Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x-axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.

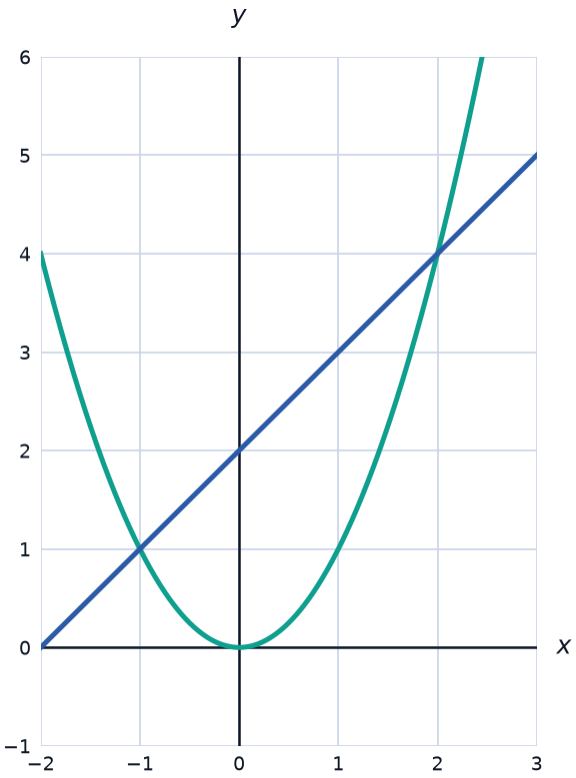
9 [Intermediate] • Target: Area between curves and a table of values

Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)]\,dx$.

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10 **[Intermediate]** • **Target: Visualizing area between two curves**

The graph shows $y = x^2$ and $y = x + 2$ from an earlier problem, together with the region between them.



11 **[Intermediate]** • **Target: A capstone area problem**

Find the area of the region bounded by $y = x^3 - x$ and $y = 0$ on $[-1, 1]$, accounting for the sign change at $x = 0$.

12 **[Intermediate]** • **Target: Area between an exponential curve and a line**

Find the area between $y = e^x$ and $y = x + 1$ on $[0, 2]$ (note $e^x \geq x + 1$ for all x, tangent only at $x = 0$).

13 **[Intermediate]** • **Target: An applied area-between-curves problem**

Two competing companies' profit rates (in thousands of dollars per year) are $P_1(t) = 50 - 2t$ and $P_2(t) = 30 + t$ over the first 10 years. Find the total difference in cumulative profit between the two companies over this period, using the area between the curves.

14 **[Intermediate]** • **Target: Intersecting Curves with Multiple Regions**

Find the area between $y = x^3$ and $y = x^6$ on $[0, 1]$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

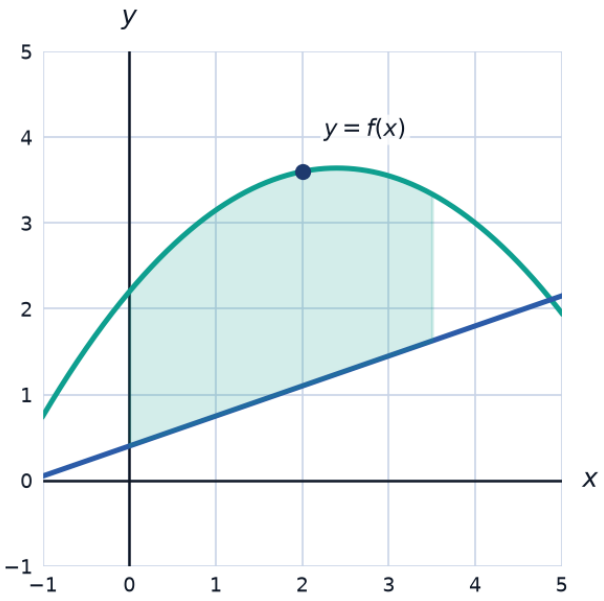
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15 [Challenging] • Target: Area Partitions & Horizontal/Vertical Lines • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Find the area between $y = x$ and $y = x^3$ on $[0, 1]$.

16 [Challenging] • Target: Area between a curve and the x-axis • Cross-Lesson: 1.01 Limits Graphically and Numerically

Find the area between $y = x$ and $y = x^4$ on $[0, 1]$.



17 [Challenging] • Target: Area between two curves (vertical strips) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Find the area between $y = x^4$ and $y = x^7$ on $[0, 1]$.

18 [Challenging] • Target: Regions requiring horizontal strips • Cross-Lesson: 2.01 The Definition of the Derivative

Find the area between $y = x^2$ and $y = x^5$ on $[0, 1]$.

19 [Challenging] • Target: Regions with multiple intersection points • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find the total area between $y = x^3 - 36x$ and the x-axis on $[-6, 6]$, accounting for the sign change at $x = -6, 0, 6$.

20 [Challenging] • Target: Area using the more efficient variable of integration • Cross-Lesson: 3.01 The Chain Rule

Find the total area between $y = x^3 - 9x$ and the x-axis on $[-3, 3]$, accounting for the sign change at $x = -3, 0, 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Area between a curve and the x-axis

Find the area between $y = x^2$ and $y = -8x - 12$.

a. $\frac{35}{3}$

b. $\frac{32}{3}$

c. 8

d. $\frac{64}{3}$

22 [Foundational] • Target: Area between two curves (vertical strips)

Find the area between $y = x^2$ and $y = 7x - 10$.

a. 9

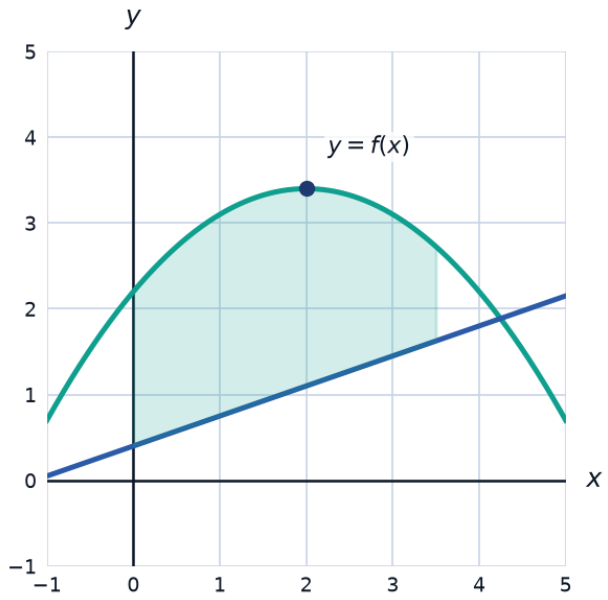
b. 5

c. $\frac{9}{2}$

d. $\frac{11}{2}$

23 [Foundational] • Target: Regions requiring horizontal strips

Find the area between $y = x^2$ and $y = -x + 6$.



a. $\frac{25}{2}$

b. $\frac{131}{6}$

c. $\frac{125}{3}$

d. $\frac{125}{6}$

24 [Foundational] • Target: Regions with multiple intersection points

Find the area between $y = x^2$ and $y = -2x$.

a. 2

b. $\frac{4}{3}$

c. $\frac{8}{3}$

d. $\frac{7}{3}$

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25 [Foundational] • Target: Area using the more efficient variable of integration

Find the area between $y = x^2$ and $y = 3x$.

a. 9

b. $\frac{11}{2}$

c. $\frac{9}{2}$

d. 4

26 [Foundational] • Target: Setting up but not evaluating (AP-style)

Find the area between $y = x^2$ and $y = 5x - 6$.

a. $\frac{1}{2}$

b. $\frac{1}{6}$

c. $\frac{1}{3}$

d. $\frac{7}{6}$

27 [Foundational] • Target: Area between curves and a table of values

Find the area between $y = x^2$ and $y = 6x$.

a. 72

b. 18

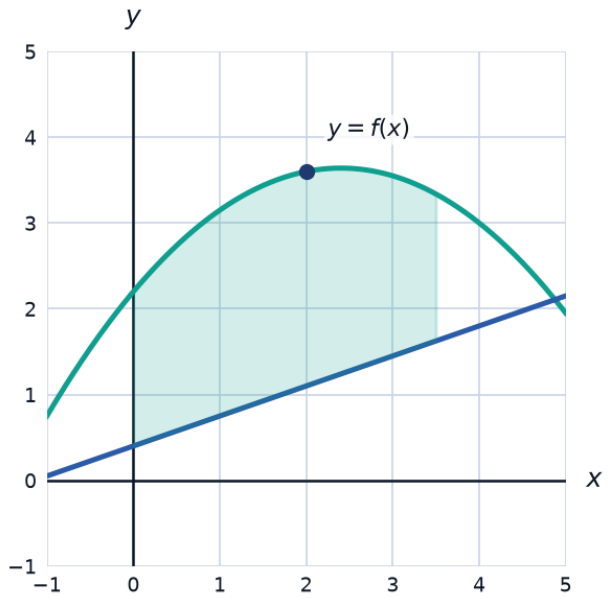
c. 37

d. 36

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28 [Foundational] • Target: Visualizing area between two curves

Find the area between $y = x^2$ and $y = 9x - 14$.



a. $\frac{25}{2}$

b. $\frac{125}{6}$

c. $\frac{131}{6}$

d. $\frac{125}{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A capstone area problem

Find the area between $y = x^4$ and $y = x^9$ on $[0, 1]$.

a. $-\frac{1}{10}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$

d. $\frac{5}{36}$

30 [Intermediate] • Target: Area between an exponential curve and a line

Find the area between $y = x^2$ and $y = x^3$ on $[0, 1]$.

a. $\frac{1}{12}$

b. $\frac{7}{12}$

c. $\frac{1}{6}$

d. $-\frac{1}{12}$

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31 [Intermediate] • Target: An applied area-between-curves problem

Find the area between $y = x^3$ and $y = x^4$ on $[0, 1]$.

a. $\frac{9}{20}$

b. $-\frac{1}{20}$

c. $\frac{1}{12}$

d. $\frac{1}{20}$

32 [Intermediate] • Target: Area Between Curves Integrated with Respect to x

Find the area between $y = x^4$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{30}$

b. $-\frac{1}{30}$

c. $\frac{1}{20}$

d. $\frac{11}{30}$

33 [Intermediate] • Target: Area Between Curves Integrated with Respect to y

Find the area between $y = x^3$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{12}$

b. $\frac{5}{12}$

c. $\frac{2}{15}$

d. $-\frac{1}{12}$

34 [Intermediate] • Target: Intersecting Curves with Multiple Regions

Find the total area between $y = x^3 - 4x$ and the x-axis on $[-2, 2]$, accounting for the sign change at $x = -2, 0, 2$.

a. 4

b. $\frac{17}{2}$

c. 10

d. 8

35 [Intermediate] • Target: Area Partitions & Horizontal/Vertical Lines

Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 3]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(1) = 15$, $h(2) = 6$, $h(3) = 10$. Use the Trapezoidal Rule with these 3 equal subintervals (width 1) to approximate the area between the curves.

a. 39

b. 29

c. 31

d. 30

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36 [Intermediate] • Target: Area between a curve and the x-axis

Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 6]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 2$, $h(2) = 12$, $h(4) = 13$, $h(6) = 2$. Use the Trapezoidal Rule with these 3 equal subintervals (width 2) to approximate the area between the curves.

a. 58

b. 56

c. 52

d. 54

37 [Intermediate] • Target: Area between two curves (vertical strips)

Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 12]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 10$, $h(4) = 15$, $h(8) = 8$, $h(12) = 8$. Use the Trapezoidal Rule with these 3 equal subintervals (width 4) to approximate the area between the curves.

a. 128

b. 132

c. 124

d. 164

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Regions requiring horizontal strips • Cross-Lesson: 2.01 The Definition of the Derivative

Find the area between $y = e^{2x}$ and $y = 2x + 1$ on $[0, 2]$ (note $e^{2x} \geq 2x + 1$ for all x, tangent only at $x = 0$), rounded to two decimal places.

a. 21.8

b. 24.96

c. 16.64

d. 20.8

39 [Challenging] • Target: Regions with multiple intersection points • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find the area between $y = e^{2x}$ and $y = 2x + 1$ on $[0, 1]$ (note $e^{2x} \geq 2x + 1$ for all x, tangent only at $x = 0$), rounded to two decimal places.

a. 2.19

b. 1.43

c. 0.96

d. 1.19

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40 [Challenging] • Target: Area Between Curves Integrated with Respect to x • Cross-Lesson: 3.01 The Chain Rule

What is the area of the region enclosed by $y = x^2$ and $y = 4$?

a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 16

d. 8

41 [Challenging] • Target: Area Between Curves Integrated with Respect to x • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Which integral represents the area enclosed by $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{4}$?

a. $\int_0^{\pi/4} (\cos x - \sin x)dx$

b. $\int_0^{\pi/4} (\sin x - \cos x)dx$

c. $\int_0^{\pi/4} (\cos x + \sin x)dx$

d. $\int_0^{\pi/2} \cos x dx$

42 [Challenging] • Target: Area Between Curves Integrated with Respect to y • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

When is it advantageous to integrate with respect to y rather than x?

a. When the boundary curves are expressed as $x = f(y)$ and solving for y requires multiple piecewise branches

b. When the area is negative

c. Always for quadratic curves

d. Never on the AP Exam

43 [Challenging] • Target: Intersecting Curves with Multiple Regions • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

The graphs of $y = x^3$ and $y = x$ enclose how many symmetric regions?

a. 2 symmetric regions

b. 1 region

c. 3 regions

d. Infinitely many

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44 [Challenging] • Target: Intersecting Curves with Multiple Regions • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Why does $\int_{-1}^{1}(x - x^3)dx = 0$ NOT represent the enclosed area?

a. The integrand is odd, causing positive and negative signed areas to cancel out

b. The curves do not intersect at 0

c. The integral diverges

d. The bounds are incorrect

45 [Challenging] • Target: Area Partitions & Horizontal/Vertical Lines • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Find the area enclosed by $y = e^x$, $y = e^{-x}$, and the vertical line $x = 1$.

a. $e + \frac{1}{e} - 2$

b. $e - 1$

c. $e + \frac{1}{e}$

d. $2e - 2$

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